Name of your Institute: **Manipal Institute of Technology**

Challenge Area: **Track 4 – Agent Teamwork**

Your Team Name: **SecondCortex Labs**

Registered Email ID of your Team Lead: suhaansyed2004@gmail.com

Project Title:

SecondCortex - A Secure Multi Agent Cognitive Memory System for Engineering Workflows

We are two engineering students who constantly build across coursework, hackathons, open source projects, and internships. Every day we switch between our IDE, GitHub, Slack, documentation, and terminal sessions.

We realized the real bottleneck in engineering work was not solving problems. It was reconstructing context after interruptions. After meetings or notifications, we often spend fifteen to twenty minutes remembering why we changed a function, what we were debugging, or which discussion influenced a deployment decision.

SecondCortex was born from that frustration. We chose AI Unlocked because Track 4 allows us to design a structured multi agent system on Azure that augments human thinking while enforcing strict security boundaries.

The Concept:

Modern engineering workflows are fragmented across tools. GitHub stores commits. Slack stores discussions. Monitoring tools store alerts. Ticketing systems store tasks. None preserve working state over time.

SecondCortex introduces **Structured Cognitive Snapshots**.

Instead of only retrieving logs and messages, SecondCortex captures lightweight contextual state at meaningful moments such as:

- Branch switches
- Pull request creation
- Deployment events
- Extended terminal sessions
- IDE file focus changes

Each snapshot stores structured metadata including active branch, open files, related tickets, recent commits, and referenced discussions. These snapshots are converted into state vectors and linked into a **Temporal Context Graph**.

When a user asks about the past, SecondCortex does not simply search logs. It reconstructs the nearest cognitive state and expands outward through graph relationships.

This enables: Intent reconstruction, Decision traceability, Timeline compression, Risk correlation

SecondCortex functions as a memory reconstruction engine rather than a document retrieval assistant.

Target Audience or Market:

SecondCortex is designed for: Software developers Site reliability engineers, Startup teams, Student builders, Engineering managers.

According to Statista 2024, the global developer population exceeds 28 million. A large percentage operate across collaborative tools daily.Even targeting a small fraction of developers in tool intensive environments represents significant opportunity.Initial focus includes developer ecosystems in India, North America, and Europe, with early adoption through student and startup communities.

SecondCortex integrates into: WhatsApp, Telegram, Slack, IDE extensions,Command line.

It embeds into existing workflows rather than adding new dashboards.

Personas:

Persona 1: On Call Engineer

An engineer investigating a production incident switches between logs, dashboards, Slack threads, and pull requests. After two hours they struggle to remember which hypothesis they already tested.

SecondCortex reconstructs the deployment timeline, related commits, incident discussions, and snapshot of debugging context, restoring clarity instantly.

Persona 2: Deep Work Developer

A developer working on a complex feature is interrupted by meetings. Returning to code requires rebuilding intent. SecondCortex restores branch context, file history, related discussions, and prior working state, enabling rapid focus recovery.

How it works:

SecondCortex implements structured agent teamwork aligned with Track 4.

Planner Agent

- Interprets user intent
- Defines temporal scope
- Generates a **directed execution graph**
- Decomposes tasks into parallel subtasks

Planner outputs must pass policy validation before execution.

SecondCortex executes subtasks in parallel where possible to reduce latency.
If a retrieval node fails or returns incomplete data, fallback nodes are triggered automatically. The Planner can regenerate subtasks without restarting the full workflow.

This makes the system resilient rather than sequential.

Retriever Agent

- Executes scoped API calls in parallel
- Uses pre approved read only OAuth scopes
- Retrieves structured cognitive snapshots

• Performs semantic vector search
• Builds the Temporal Context Graph linking commits, discussions, deployments, and snapshots

All actions are logged and permission validated.

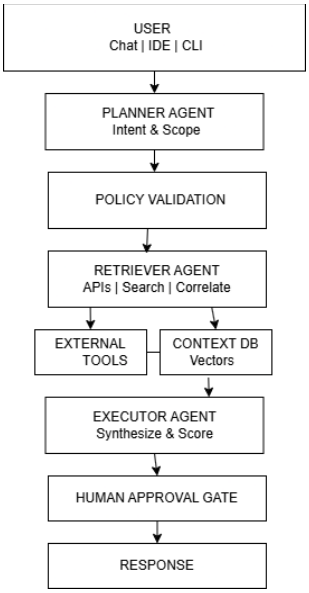
Executor Agent and Validation Loop

The Executor Agent does not immediately finalize output.

After generating a structured timeline:

• It produces an initial reasoning summary
• A secondary reasoning pass re evaluates correlation confidence
• Discrepancies are flagged
• A confidence score is attached to the result

This internal validation loop reduces hallucination risk and strengthens reliability. Only after this reasoning validation does the system present output to the user. This demonstrates cooperative agent behavior rather than linear execution.



Multi Step Orchestration Example

User asks: Why was the payment service rolled back last Friday?

Planner:

- Identifies repository and time window
- Generates execution graph

Retriever:

- Fetches deployment logs
- Retrieves Slack incident thread
- Pulls linked ticket
- Collects related commits
- Retrieves nearest cognitive snapshot
- Builds correlation graph

Executor:

- Constructs root cause timeline
- Identifies responsible change
- Summarizes discussion
- Applies risk model

Returns recommendation with confidence score

This demonstrates coordinated multi agent reasoning.

Security Architecture:

SecondCortex is built on enforceable constrained agency.

Security layers include:

• Planner outputs must pass policy middleware before execution
• Retriever can only access pre approved read only scopes
• Executor cannot call external systems directly
• Azure Active Directory enforces role based boundaries
• Azure Container Apps isolate each deployment environment
• Immutable audit logs record every agent decision
• No arbitrary code execution pathways exist

Agents operate inside strict boundaries. No agent can escalate privileges or bypass policy validation.

SecondCortex is designed for cognitive assistance, not infrastructure control.

Core Technologies:

- **Python with FastAPI for orchestration**
- **Azure OpenAI for reasoning and embeddings**
- **Azure Container Apps for isolated deployment**
- **Azure Active Directory for identity and role enforcement**
- **Vector database for semantic retrieval**
- **Structured cognitive snapshot engine**
- **Messaging integrations**

Architecture follows scalable multi agent orchestration patterns compatible with Microsoft's agent frameworks.

The Business Plan:

Freemium model:

Free tier with limited integrations and local deployment.
Professional tier with team features and cloud synchronization.
Enterprise tier with advanced policy controls, audit logs, and single sign on.

Initial adoption through student communities and startup teams, followed by expansion into enterprise environments.

Additional Information:

SecondCortex reduces engineering time lost to context switching without increasing system risk. It demonstrates structured agent teamwork, secure orchestration, and scalable deployment using **Microsoft Azure**. It augments engineers instead of replacing them.